

Metric rigidity at the boundary of cubic compactifications

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Abstract

We deduce a sharp Cartier-index criterion for complex-structure rigidity on singular Kähler–Einstein Fano varieties. If $-K_X = rL$ with L ample Cartier and $r > \dim(X)/2 + 1$, then the tangent sheaf is stable and every parallel orthogonal complex structure on the regular locus equals the given one or its negative. The proof combines the decomposition theorem of Druel–Guenancia–Păun with a Hilbert-polynomial index bound and local positive-Ricci curvature. Applied to cubic n -fold limits, $n \geq 5$, it identifies the ordinary metric Gromov–Hausdorff compactification with cubic GIT modulo conjugation, including the singular boundary. We also record the complex-preserving comparison as an attributed consequence of the unrestricted ordinary-double-point gap, Li–Liu’s global Reeb minimum, and Spotti–Sun’s established transfer. The complex algebraic comparison is already publicly stated in stronger form by Kong–Shen–Zhao–Zhao; no first-solution claim is made.

Status and provenance. The upstream manuscript [1] is pinned at repository commit `adc7f1241b42` (the full hash is in the source manifest). Its theorem is a principal external input, not a theorem reproved in this note. Separate scoped adversarial audits inspected its geometric and semigroup arguments and its sole fourfold companion. These are evidence, not formal certification. No Lean verification of the gap or these consequences is claimed. AI tools were used extensively in research, drafting, and verification. This note has not undergone conventional human peer review. The additional assertion concerns singular metric rigidity and the fibers of the forgetful map. The splitting principle, the conjugation convention, and the complex cubic comparison are credited to their original sources.

1 Exact target and dependency diagram

Fix $n \geq 5$ and let

$$G_n = \mathbb{P}(H^0(\mathbb{P}^{n+1}, \mathcal{O}(3))) // \mathrm{SL}(n+2).$$

The complex GH moduli object retains complex structure: its points are biholomorphic isometry classes of Kähler–Einstein cubic n -folds, with $\mathrm{Ric}(\omega) = \omega$, and its compactification is the complex/polarized GH compactification used in [3]. Bare metric-space isometry forgets complex structure. Write $\overline{\mathcal{M}}_n^{\mathrm{GH}, \mathrm{metric}}$ for the closure of the underlying compact metric spaces of smooth KE cubics, with ordinary metric GH topology. Write σ for coefficientwise conjugation on $G_n(\mathbb{C})$. We prove that conjugation orbits are exactly the fibers of the forgetful map on the entire boundary.

The good moduli space on the K side has closed complex points representing K-polystable $\mathbb{Q}$ -Fano varieties that admit a $\mathbb{Q}$ -Gorenstein deformation to a smooth cubic. Equivalently one

considers the corresponding closure of the smooth cubic locus in the smoothable K-moduli space, with its complex analytic topology. Equal dimension and volume alone do not select that smoothing locus. The established smoothable KE/GH correspondence is the relevant identification with these closed points [7].

The dependency diagram for the proposed deduction is

$$\begin{array}{c}
\widehat{\text{vol}}(x, X) \leq 2(k-1)^k \quad (2 \leq k \leq n) \quad [1] \\
\Downarrow \quad \text{global Reeb minimum and density identity [2]} \\
\Theta(C) \leq 2(1 - 1/k)^k \quad \text{for isolated singular metric cones} \\
\Downarrow \quad \text{iteration and Cartier-root theorem [3]} \\
-K_Z = (n-1)L_Z, \quad L_Z \text{ Cartier}, \quad L_Z^n = 3 \\
\Downarrow \quad \text{Fujita classification; CM positivity; continuity [6, 3]} \\
\overline{\mathcal{M}}_n^{\text{GH}, \mathbb{C}} \cong_{\text{homeo}} G_n(\mathbb{C}).
\end{array}$$

Only the upper gap bound is needed; its equality classification is not used. No scheme/stack isomorphism, equivalence of moduli functors, or assertion about every nonclosed semistable orbit is deduced here.

Theorem 1 (Sharp Cartier-index rigidity). *Let X be an irreducible normal projective klt Fano of dimension $m \geq 1$ carrying a weak KE current with bounded potentials, smooth on X_{reg} . Suppose an ample Cartier line bundle L and a positive integer r satisfy the actual line-bundle identity $\omega_X^{[-1]} \simeq L^{\otimes r}$ and $r > m/2 + 1$. Then T_X , and the tangent sheaf of every finite quasi-étale cover, is slope stable for the anticanonical polarization. No such cover is a product of two positive-dimensional normal projective varieties. Every parallel orthogonal complex structure on (X_{reg}, g) is J or $-J$. The strict numerical threshold is sharp in this class.*

Theorem 2 (Cubic metric boundary). *For $n \geq 5$, using the upper gap theorem of [1], there are natural homeomorphisms*

$$\overline{\mathcal{M}}_n^{\text{GH}, \mathbb{C}} \simeq G_n(\mathbb{C}), \quad \overline{\mathcal{M}}_n^{\text{GH}, \text{metric}} \simeq G_n(\mathbb{C})/\langle \sigma \rangle.$$

The first space has the closed-point interpretation in the cubic smoothable K-polystable coarse space described above. Every metric isometry between its KE boundary varieties is holomorphic or anti-holomorphic; in the latter case it is a holomorphic isomorphism to the conjugate variety. In either formulation the cubic hyperplane Cartier line bundle is preserved.

The first comparison uses the displayed established transfer. The additional argument for the second comparison is

$$\begin{aligned}
&\text{singular KE stable-factor cover [12]} \implies \dim(Y_i) \geq r-1 \implies T_X \text{ stable} \\
&\implies I = \pm J \implies \text{all metric fibers are conjugation orbits} \\
&\implies \text{compact-Hausdorff quotient homeomorphism.}
\end{aligned}$$

The inequality is about a genuine Cartier root, not an arbitrary $\mathbb{Q}$ -Cartier divisor. The metric isometry statement uses the regular-locus theorem for the actual GH limits; no claim about arbitrary metric completions of nonsmoothable weak KE varieties is needed.

2 Cartier index and parallel complex structures

Proof of Theorem 1. Druel–Guenancia–Păun [12, Theorem A] supply a finite quasi-étale cover $f : Y \rightarrow X$ and an isometric algebraic product of positive-dimensional weak KE klt Fanos Y_i with stable tangent sheaves. Their theorem treats the incomplete regular locus; we do not apply complete de Rham splitting there. Put $M = f^*L$. Quasi-étaleness gives $\omega_Y^{[-1]} \simeq M^{\otimes r}$.

For each factor choose a regular point b of the complementary product. The restriction $M_i = M|_{Y_i \times \{b\}}$ is ample Cartier. On $Y_{i,\text{reg}} \times \{b\}$ the smooth product canonical formula identifies ω_Y with ω_{Y_i} tensored with the fixed one-dimensional space $\det(T_b^*B)$. Normal reflexive extension therefore gives $\omega_{Y_i}^{[-1]} \simeq M_i^{\otimes r}$ globally. This also works for the factors of any algebraic product cover: quasi-étale pullback is klt, and adjoining smooth coordinates tests klt on a factor.

Let Z be such a factor, of dimension d , with $-K_Z = rA$ and A ample Cartier. For $j = 1, \dots, r-1$, klt Kawamata–Viehweg vanishing applies to $-jA$, since $-jA - K_Z = (r-j)A$ is ample [18, Theorem 2.22]. Negative ampleness gives $H^0(Z, -jA) = 0$ as well. Hence the nonzero degree- d polynomial $P_Z(t) = \chi(Z, A^{\otimes t})$ has the $r-1$ distinct roots $-1, \dots, -(r-1)$, so $d \geq r-1$. Euler characteristic is polynomial at every integer, including negative integers [15, Lemma 33.45.1]. Two factors would imply $m \geq 2(r-1)$, contradicting the strict index bound. The DGP cover therefore has one factor and T_Y is stable.

Stability descends without a $\mathbb{Q}$ -factoriality assumption. If a proper saturated $\mathcal{F} \subset T_X$ had slope at least $\mu(T_X)$, pull it back on the finite étale big open set and extend reflexively and saturate to $\mathcal{G} \subset T_Y$. Both complements have codimension at least two. Choose a general complete-intersection curve of large anticanonical degree in that big open set. On its étale inverse image, ordinary bundle degrees multiply by $\deg f$, giving

$$\mu_{f^*(-K_X)}(\mathcal{G}) = \deg(f)\mu_{-K_X}(\mathcal{F}), \quad \mu_{-K_Y}(T_Y) = \deg(f)\mu_{-K_X}(T_X).$$

This contradicts stability upstairs. The same hypotheses hold on every finite quasi-étale cover, proving its tangent stability as well.

Let I be another parallel orthogonal complex structure on X_{reg} . The real two-form $\eta(u, v) = g(Iu, v)$ is parallel. Its $(2, 0)$ part relative to J is parallel. On a KE manifold with $\text{Ric}(g) = \lambda g$, $\lambda > 0$, contraction of Chern curvature on $\Lambda^p T^{*(1,0)}$ is $-p\lambda \text{Id}$, up to the common curvature sign convention. A parallel section has zero curvature action, so this contraction annihilates it. Thus no nonzero parallel $(2, 0)$ part exists. This is a local identity, requiring no completeness or integration.

Consequently η is of type $(1, 1)$, equivalently $-JIJ = I$, so $IJ = JI$. The parallel endomorphism I of $T_{X_{\text{reg}}}^{1,0}$ is holomorphic. The reflexive sheaf $\mathcal{H}om(T_X, T_X)$ extends it across the codimension-two complement; GAGA makes it algebraic [19, Theorem 2]. A stable sheaf is simple: the kernel/image slope inequalities exclude a nonzero endomorphism of intermediate generic rank. Characteristic polynomial coefficients extend to global regular functions on normal projective X , hence are constant. Subtracting a complex eigenvalue then gives an endomorphism of deficient generic rank, which must be zero. Thus $I = c \text{Id}$. The equation $I^2 = -1$ gives $c = \pm i$, or $I = \pm J$ as a real endomorphism.

For sharpness take $X = \mathbb{P}^{r-1} \times \mathbb{P}^{r-1}$, $r \geq 2$, and $L = \mathcal{O}(1, 1)$. Then $m = 2r - 2$ and $-K_X = rL$. The product of equally normalized Fubini–Study metrics is KE. Conjugation on only one factor is an isometry whose pullback complex structure is $(-J_1, J_2)$, neither J nor $-J$. This falsifies including equality in the uniform criterion; it is not a cubic counterexample. $\square$

3 The infimum-to-metric implication

Write $\widehat{\text{vol}}(x, X) = \inf_v A_X(v)^k \text{vol}(v)$ for a boundary-zero complex algebraic klt germ of dimension k . Ordinary volume is normalized by $k!$ times the leading valuation-ideal colength.

Proposition 3. *Assume the algebraic gap bound in dimension $k \geq 2$. Then an isolated singular Ricci-flat Kähler cone with compact smooth Sasaki–Einstein link has density at most $b_k = 2(1 - 1/k)^k$.*

Proof. Use the cone metric $dr^2 + r^2 g_Y$, with

$$\Theta(C(Y)) = \frac{\text{Vol}(Y)}{\text{Vol}(S^{2k-1})}.$$

The Sasakian cone has a normal affine algebraic completion; a quasi-regular CR-preserving approximation constructs it as an orbifold line-bundle cone [4, §3]. It is $\mathbb{Q}$ -Gorenstein. The positive pluricanonical Reeb weight gives klt, by [5, Lemma 6.1]. Rationality and $\mathbb{Q}$ -Gorensteinness alone are not being substituted for this klt criterion. The cone germ has zero boundary.

There is a convenient way to respect the genuine top-form hypothesis in the setup of Li–Liu §6. If $\pi_1(Y)$ is nontrivial, Myers’ theorem gives a finite universal covering of degree $d \geq 2$. Bishop’s inequality on that cover yields $\Theta(C(Y)) \leq 1/d \leq 1/2 \leq b_k$. If Y is simply connected, the Ricci-flat punctured cone has a global parallel holomorphic top form. Its Euler weight is k ; it extends reflexively across the vertex and is equivariant for the cone torus. This satisfies the canonical-form setup of Li–Liu.

Li–Liu’s Theorem 1.9 (=6.2) states that the metric Reeb valuation v_ξ globally minimizes over *all* real valuations centered at the vertex. The theorem includes irregular Reeb fields. Their Lemma 6.8 and limiting proof use quasi-regular structures converging smoothly; these approximants are not incorrectly assumed to remain Einstein. Lemma 6.3 and the same limiting argument give

$$A_X(v_\xi) = k, \quad \widehat{\text{vol}}_X(v_\xi) = k^k \Theta(C(Y)).$$

The ratio to the round sphere cancels the fixed conversion between contact volume and Riemannian link volume. Both sides are continuous in the quasi-regular approximation. Thus, in the simply connected case,

$$k^k \Theta(C(Y)) = \widehat{\text{vol}}(o, X) \leq 2(k-1)^k.$$

An upper bound on the infimum alone would not imply this inequality for an arbitrary Reeb valuation. The global minimum is essential. $\square$

Flat $\mathbb{C}^k$ has density 1 and normalized volume k^k ; it is excluded from the singular assertion. The ODP degree valuation has discrepancy $k-1$ and volume 2, consistent with the stated normalization. No direct application of Li–Liu to a cone with singular link is made.

4 From density to polarized cubic limits

Spotti–Sun’s constant $A'(n)$ is the supremum of the densities of products of a flat factor with an isolated singular cone having smooth link. Their iterated-cone argument treats other singular links [3, Lemma 5.3 and Theorem 5.2]. The required transverse dimensions are exactly $k = 2, \dots, n$. A normal klt algebraic curve is smooth, and the relevant KE singular strata have complex codimension at least two.

The sequence b_k increases. For real $t > 1$,

$$\frac{d}{dt} \log b_t = \log(1 - 1/t) + \frac{1}{t-1} = u - \log(1 + u) > 0, \quad u = \frac{1}{t-1}.$$

Products with Euclidean space do not change density, so the preceding proposition gives $A'(n) \leq b_n$.

For a smooth cubic, adjunction gives

$$-K_X = (n-1)H, \quad H^n = 3, \quad V = (-K_X)^n = 3(n-1)^n.$$

Spotti–Sun Theorem 5.2 requires

$$V > \frac{1}{2}A'(n)(n+1)^n.$$

This follows strictly from

$$\frac{1}{2}b_n(n+1)^n = (n-1)^n(1+1/n)^n < e(n-1)^n < 3(n-1)^n.$$

Its conclusion supplies both Gorenstein canonical singularities and the Cartier root L_Z with $-K_Z = (n-1)L_Z$. Preserving canonical singularities alone would not preserve the required polarization. The theorem’s proof uses the polarized smoothability of the limit, the tangent-cone Cartier index comparison, and quotient-smoothing/rigidity arguments. These are inherited hypotheses and conclusions, not consequences of density alone for arbitrary $\mathbb{Q}$ -Fanos.

Fujita’s Gorenstein del Pezzo classification applies to (Z, L_Z) [6]. Its intermediate-cohomology hypothesis holds for all integer twists: Kawamata–Viehweg vanishing applies to tL_Z for $t \geq 2 - n$, since $tL_Z - K_Z = (t + n - 1)L_Z$ is ample. Serre duality applies the same argument at twist $-t - (n - 1)$ for $t \leq -1$. These ranges cover all integers for $n \geq 3$, and klt makes Z Cohen–Macaulay. The degree-three case has $h^0(Z, L_Z) = n + 2$ and L_Z very ample, so it embeds Z as a cubic hypersurface in $\mathbb{P}^{n+1}$. This uses the canonical Gorenstein version, without imposing an extra terminality assumption.

5 Moduli continuity and the precise consequence

Here is an explicit positivity check for the relevant CM polarization. Put $S = \mathbb{P}(H^0(\mathbb{P}^{n+1}, \mathcal{O}(3)))$, let h and s denote the hyperplane classes on $\mathbb{P}^{n+1}$ and S , and put $r = n - 1$. The universal cubic has class $3h + s$, and its relative anticanonical class is $rh - s$. In the standard Fano normalization,

$$\begin{aligned} c_1(\lambda_{\text{CM}}) &= -\pi_*((rh - s)^{n+1}(3h + s)) \\ &= r^n(3(n+1) - r)s = 2(n+2)(n-1)^n s. \end{aligned}$$

The expression is computed on the product; the universal-cubic factor implements pushforward from the hypersurface. Any positive conventional rescaling preserves the sign. The coefficient is strictly positive, so the CM polarization agrees in sign with the standard GIT polarization.

Spotti–Sun §5.2 applies their moduli continuity method from §4.2. The initial Fermat cubic is KE; the smooth cubic moduli locus is connected. KE existence is open there, and the cubic-limit and CM facts make it closed. The map from the complex GH compactification to GIT is continuous; uniqueness of the KE metric on a fixed Fano variety gives injectivity. Its compact image contains the dense smooth GIT locus, giving surjectivity. Compactness and the Hausdorff analytic GIT topology give a homeomorphism. This is precisely the established transfer [3, Theorem 1.3(2)], rather than a new continuity theorem.

Corollary 4 (Audited conditional consequence). *If the unrestricted algebraic normalized-volume gap holds in every dimension $2 \leq k \leq n$, then for $n \geq 5$ the complex GH compactification of smooth KE cubic n -folds is naturally homeomorphic to $G_n(\mathbb{C})$. Its closed points also identify with the corresponding smoothable K -polystable Fano limits in the cubic smoothing locus.*

Taking the upper-bound theorem of [1] as the external input removes the dimension-by-dimension metric-gap hypothesis in [3, Theorem 1.3(2)]. This is the inherited complex-moduli deduction. The topological result alone does not upgrade this deduction to schemes, stacks, or nonclosed semistable points.

6 Isometries on the singular cubic boundary

Proof of Theorem 2. The complex comparison is the preceding corollary with [1]. Its KE boundary varieties are normal klt cubic hypersurfaces, and adjunction gives $-K_X = (n-1)H_X$ with H_X ample Cartier. Since $n-1 > n/2 + 1$ for $n \geq 5$, Theorem 1 applies.

Donaldson–Sun [13, Proposition 2.14] identify the metric singular set and the analytic singular set in this GH-limit setting. A metric isometry $F : X \rightarrow Z$ therefore maps X_{reg} onto Z_{reg} . Myers–Steenrod makes it a smooth Riemannian isometry locally there; this local assertion requires no completeness of the regular locus. Thus $I = F^*J_Z$ is parallel and orthogonal. Theorem 1 gives $I = J_X$ or $-J_X$ with a single global sign. Hence F is holomorphic on the regular locus, or holomorphic to $\bar{Z}$.

To extend it, choose an analytic chart about $F(x)$ and a neighborhood of x whose continuous image lies in that chart. Each target coordinate composed with F is holomorphic on the regular locus and locally bounded. Normal Riemann extension makes these functions holomorphic on the whole neighborhood. They equal the continuous original coordinates by density and satisfy the target equations. Repeat for F^{-1} . This gives a global holomorphic isomorphism, algebraic by GAGA [19, Proposition 15], with target Z or $\bar{Z}$. The metric and analytic topologies agree in the GH algebraization; the local extension principle also appears in [13, Proposition 2.4].

Grothendieck’s Picard Lefschetz theorem [14, Exposé XII, Corollaire 3.7] applies to global projective complete intersections of dimension at least three, including singular ones. It gives $\text{Pic}(X) = \mathbb{Z}[H_X]$. Canonical preservation implies $(n-1)(F^*H_Z - H_X) = 0$, taking $\bar{Z}$ in the antiholomorphic case. The Picard group is torsion free, so $F^*H_Z = H_X$. The complete hyperplane system is the cubic embedding, since the hypersurface sequence gives $h^0(X, H_X) = n+2$. Accordingly these isomorphisms give precisely the corresponding projective cubic GIT points and their conjugates.

The forgetful map from the complex GH compactification to ordinary metric GH space is continuous. Any metric limit of smooth KE cubics has a polarized subsequential lift by polarized compactness, so its image is exactly $\overline{\mathcal{M}}_n^{\text{GH,metric}}$. The preceding isometry classification makes its fibers σ -orbits. Conversely complex conjugation changes both J and the real Kähler two-form by sign, preserving the Riemannian metric. The induced map from $G_n(\mathbb{C})/\langle\sigma\rangle$ is therefore a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. $\square$

No effective GH-distance estimate or modulus of continuity is asserted. The sharpness example in Theorem 1 concerns general Fanos; it does not establish a failure of cubic metric rigidity in dimension four.

7 Prior results and the additional deduction

The lower-dimensional comparison is prior work: cubic surfaces by Odaka–Spotti–Sun, cubic threefolds by Liu–Xu, and cubic fourfolds by Liu [8, 9, 10]. Cubic curves are not included in a positive-KE Fano headline. Smooth KE existence should not be advertised as newly discovered here.

The primary author page [11] currently links a September 30, 2026 manuscript, publicly downloaded on October 6, 2026 (Pacific time). Its Theorem 1.1 states the arbitrary-dimensional cubic comparison, including stacks and good moduli spaces; its proof uses refined local inequalities and tangent-sheaf semistability. The actual earliest web posting is unverified, and a manuscript date alone does not establish public priority. Its present public availability is verified. This audit does not independently certify every lemma of that separate paper.

The complex cubic comparison is therefore prior public work. Our gap-based account instantiates the known Spotti–Sun reduction; the unrestricted gap proof belongs to OpenAI, and global Reeb minimization to Li–Liu.

Theorem 1 and the singular metric-fiber argument are the additional deductions recorded here. Smooth cubic tangent stability already follows from Peternell–Wiśniewski [16, Corollary 1.5 and Theorem 2.3]. Smooth positive-KE rigidity up to factorwise conjugation is explained by Spotti [17, Theorem 1.1.1]. Odaka–Spotti–Sun expressly give the ordinary metric quotient by conjugation for their singular del Pezzo-surface compactifications [8, §§1–2]. The present argument checks the singular higher-dimensional boundary by the Cartier-index exclusion of product covers, reflexive endomorphisms, and the GH regular-locus theorem. It uses DGP’s splitting, not a newly invented decomposition principle. The threshold’s product counterexample is elementary.

Primary-source comparisons distinguish these exact scopes; they do not certify first public disclosure of every possible equivalent corollary. We make no first-priority claim and do not advertise a new solution of the complex cubic K/GIT problem. The contribution is a modest, explicitly attributed singular metric-rigidity criterion and its boundary application.

References

- [1] OpenAI, *The ordinary-double-point gap in every dimension*, OpenAI Math Release preprint, September 24, 2026. [Pinned manuscript](#). Sole companion: *The normalized-volume gap in dimension four*.
- [2] C. Li and Y. Liu, *Kähler–Einstein metrics and volume minimization*, Adv. Math. 341 (2019), 440–492, [arXiv:1602.05094v3](#).
- [3] C. Spotti and S. Sun, *Explicit Gromov–Hausdorff compactifications of moduli spaces of Kähler–Einstein Fano manifolds*, Pure Appl. Math. Q. 13 (2017), 477–515, [arXiv:1705.00377v1](#).
- [4] C. van Coevering, *Ricci-flat Kähler metrics on crepant resolutions of Kähler cones*, [arXiv:0806.3728v3](#).
- [5] T. C. Collins and G. Székelyhidi, *Sasaki–Einstein metrics and K-stability*, Geom. Topol. 23 (2019), 1339–1413, [doi:10.2140/gt.2019.23.1339](#).
- [6] T. Fujita, *On singular del Pezzo varieties*, in *Algebraic Geometry*, Lecture Notes in Math. 1417, Springer, 1990, 117–128, [doi:10.1007/BFb0083337](#).
- [7] C. Li, X. Wang and C. Xu, *On the proper moduli spaces of smoothable Kähler–Einstein Fano varieties*, Duke Math. J. 168 (2019), 1387–1459, [arXiv:1411.0761v4](#).

- [8] Y. Odaka, C. Spotti and S. Sun, *Compact moduli spaces of del Pezzo surfaces and Kähler–Einstein metrics*, J. Differential Geom. 102 (2016), 127–172, [arXiv:1210.0858](#).
- [9] Y. Liu and C. Xu, *K-stability of cubic threefolds*, Duke Math. J. 168 (2019), 2029–2073, [arXiv:1706.01933](#).
- [10] Y. Liu, *K-stability of cubic fourfolds*, J. Reine Angew. Math. 786 (2022), 55–77, [arXiv:2007.14320v2](#).
- [11] B. Kong, Y. Shen, J. Zhao and M. Zhao, *K-moduli of cubic hypersurfaces*, manuscript dated September 30, 2026, [primary author page](#), [public PDF](#). Access checked October 6, 2026 Pacific time.
- [12] S. Druel, H. Guenancia and M. Păun, *A decomposition theorem for $\mathbb{Q}$ -Fano Kähler–Einstein varieties*, Comptes Rendus Math. 362, Special issue S1 (2024), 93–118, [doi:10.5802/crmath.612](#).
- [13] S. Donaldson and S. Sun, *Gromov–Hausdorff limits of Kähler manifolds and algebraic geometry, II*, J. Differential Geom. 107 (2017), 327–371, [arXiv:1507.05082v1](#).
- [14] A. Grothendieck, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)*, North-Holland, 1968; corrected edition by M. Raynaud, [arXiv:math/0511279](#).
- [15] The Stacks Project Authors, *The Stacks Project*, [tag 0BEM](#), Lemma 33.45.1; [tag 0BEL](#), Section 33.45. Accessed October 6, 2026 Pacific time.
- [16] T. Peternell and J. A. Wiśniewski, *On stability of tangent bundles of Fano manifolds with $b_2 = 1$* , J. Algebraic Geom. 4 (1995), 363–384, [arXiv:alg-geom/9306010v2](#).
- [17] C. Spotti, *Degenerations of Kähler–Einstein Fano manifolds*, doctoral thesis, 2012, [arXiv:1211.5334](#).
- [18] C. Hacon, *Math 7800 notes*, 2018, [primary notes](#), Theorem 2.22 (klt Kawamata–Viehweg vanishing).
- [19] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier 6 (1956), 1–42, [primary article](#).